

Salvation Exam (*Exam Extraordinario*)

Course Title: **Inglés IX**
Term: **26-3**
Instructor: **Marron**

General Instructions: Print your name in the upper right-hand corner of this paper. Read each item carefully. Be sure that you understand exactly what is being asked of you. Be sure to write your name on any and all extra sheets of paper that are used.

Tasks will be evaluated by sub-tasks. Three (3) points are available for each sub-task: Accuracy (1 pt), Completeness (1 pt), and Sufficiency (1 pt). Points will be awarded in 0.1 increments. Answer in English unless requested to do so otherwise.

Task 1 (6 pts)

At present, AI models such as ChatGPT operate as Large Language Models (LLMs) built using transformer architecture from artificial neural networks powered by perceptrons.

1. Why are LLMs called “large language models”?
2. What is a perceptron and how (basically) does it function?
3. What is it that LLMs actually do?
4. What is Humanity's Last Exam and why is it important?

Task 2 (6 pts)

Many see AI playing an evermore important role in document translation, including opening up new opportunities for human translators¹. If true, human translators will need to develop a variety of new skills and be conversant with new concepts and procedures.

1. Access to cloud-based LLM translation is typically done through an Application Programmer Interface (API) using Python. Briefly explain i) the nature of an API, and ii) the nature of Python.
2. Briefly explain the concept of prompt engineering.
3. Metrics for the quality of machine translation are rapidly under development. Two common metrics currently in use are the bilingual evaluation understudy (BLEU) metric and the metric for evaluation of translation with explicit ordering (METEOR). Select one of these metrics and briefly explain how it works.

¹ Sundberg and Holmstrom (2024). Innovating by prompting: How to facilitate innovation in the age of generative AI, *Business Horizons*, 67 (5)

Task 3 (6 pts)

Paul Ford's recent essay in the New York Times notes that "vibe coding" is rapidly displacing professionals who have typically used coding scripts through an Application Programmer Interface (API) to complete complicated projects and create various computer-based applications².

1. What is meant by the term "vibe coding" and explain how this is rapidly changing the world of professional coding.
2. How might you use "vibe coding" in the future?

Task 4 (6 pts)

The report, AI 2027³ paints a highly dystopian future resulting from, in grand measure, alignment problems and uncontrollable emergent properties that appear in massive AI systems. In a recent essay in the New York Times Ezra Klein points out that the weaponized use of AI is becoming a terrifying reality and that strong, legal oversight is needed if we hope to avoid misuse of AI by the national security state⁴. In contrast Richard David Hames states that "Artificial intelligence is not merely a machine. It's a mirror, a partner, and perhaps a successor. It compels us to rethink reality, identity, and purpose."⁵

1. What is meant by the term, "alignment" and why must AI agents be aligned?
2. What is meant by the term, "emergent property" and why might emergent properties be worrisome in AI models?
3. Richard David Hames offers perhaps fantastical possibilities for positive interactions between AI and humans. List one important take-away from his opinion piece.

Task 5 (6 pts)

Write a short synopsis of the complete novel, *Bless Me Ultima* and briefly discuss the following:

1. The transformation of Antonio under the tutelage of Ultima.
2. Anaya's use of Native American traditions and the supernatural.
3. Your reasoned opinion of the novel.

2 Ford (2006). The A.I. Disruption We've Been Waiting for Has Arrived, New York Times Guest Opinion, Feb. 18, 2026.
3 AI Futures Project (2025). AI 2027, Accessed at <https://ai-2027.com/> on July 23, 2025
4 Klein (2006). The Future We Feared Is Already Here, New York Times Guest Opinion, Mar. 8, 2026.
5 Hames (2025). The Future of AI: A Radical Vision for Humanity's Next Chapter, Aug 19, 2025 at <https://richarddavidhames.substack.com>